CS 569 Machine Learning

Homework 3: Multiple Linear Regression with Advanced Extensions

1 Objective

In this assignment, you will build, evaluate, and improve a regression model based on multiple features. You will:

- Start with a basic Multiple Linear Regression.
- Extend it with Polynomial Features.
- Apply Ridge Regression for regularization.
- Visualize results and analyze feature impact.

2 Dataset

House price regression

This dataset includes 1,000 samples with the following columns:

- Features: Square_Footage, Num_Bedrooms, Num_Bathrooms, Year_Built, Lot_Size, Garage_Size, Neighborhood_Quality.
- Target: House_Price.

3 Tasks

3.1 Data Preparation

- Use pandas to load the dataset (as what you have done in previous assignments).
- Separate the features (X) and labels (y = House_Price).
- Use train_test_split() to split the dataset (80% train / 20% test). All results presented in this file were generated with random_state set to 42 to ensure reproducibility.

3.2 Part A: Basic Multiple Linear Regression

- Train a Multiple Linear Regression model using all features.
- Make predictions on the test set.
- Evaluate the model, as shown in Figure 1:
  - R^2
  - Mean Squared Error (MSE)
  - Model coefficients & intercept
- Visualize:
  - Actual vs. Predicted scatter plot, as shown in Figure 2.
  - Residual plot (actual minus predicted vs. actual), as shown in Figure 3.
  - Fitted line for different features, as shown in Figure 4.
- Compute and display Permutation Feature Importance for the basic linear regression, as shown in Figure 5.

3.3 Part B: Polynomial Regression

- Extend your model to use Polynomial Features with degree = 5 (or more) for all predictors.
- Train a Polynomial Linear Regression on the transformed features.
- Make predictions and evaluate R^2 and MSE.
- Visualize:
  - Actual vs. Predicted scatter plot.
  - Residual plot.
  - Fitted line for different features

3.4 Part C: Ridge Regression with Polynomial Features

- Apply Ridge Regression using the same polynomial features.
- Train the Ridge model.
- Make predictions and evaluate R^2 and MSE.
- Visualize:
  - Actual vs. Predicted scatter plot.
  - Residual plot.
  - Fitted line for different features

4 Items to submit

A zip file that contains your Python source code and a PDF report.

```
=== Part A: Basic Linear Regression ===
R²: 0.9984263636823408
MSE: 101434798.50566845
Coefficients: [  199.5132451  10225.20442447  8208.43477773   993.53717106
 14885.38441462  5146.14838284   115.06859524]
Intercept: -2010567.4202702064
```

Figure 1: Evaluation metrics

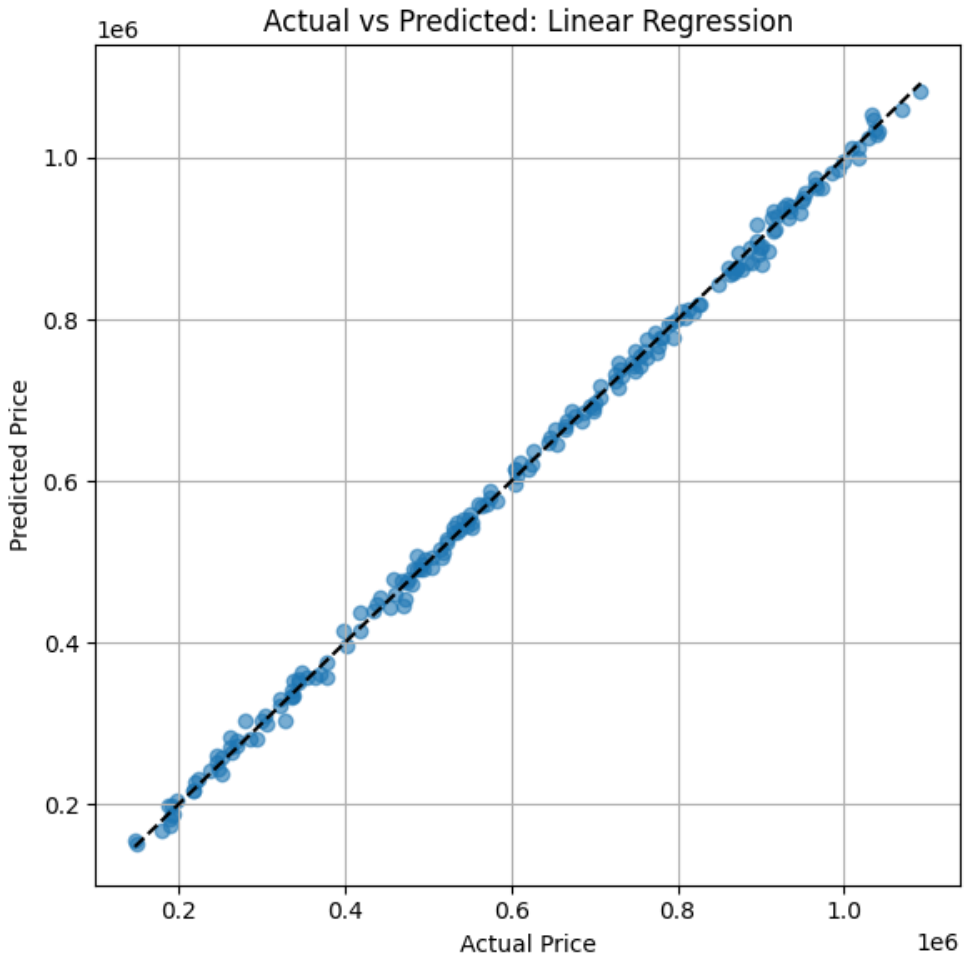

Figure 2: Actual VS. Predicted

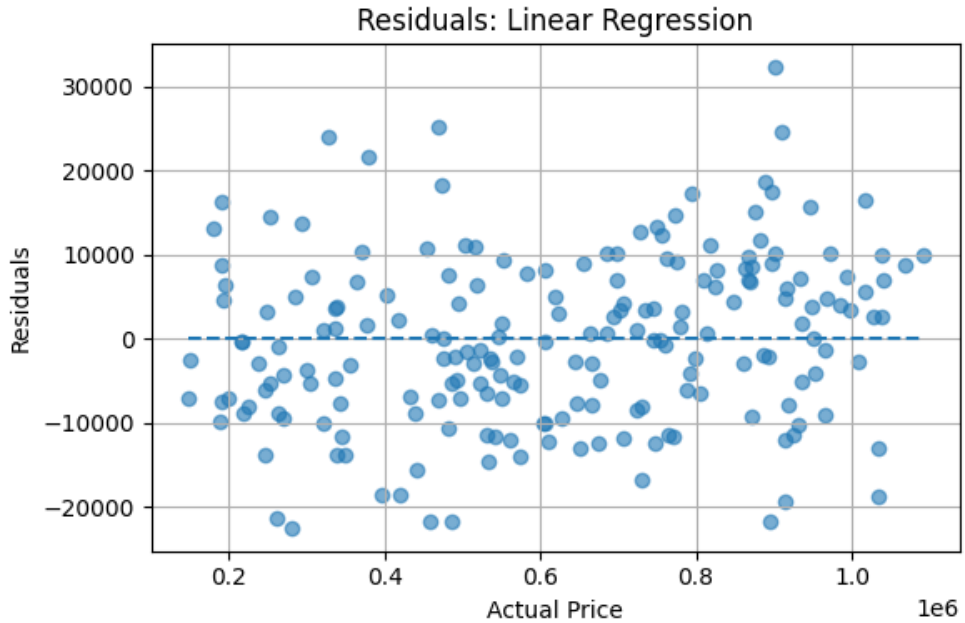

Figure 3: Residuals

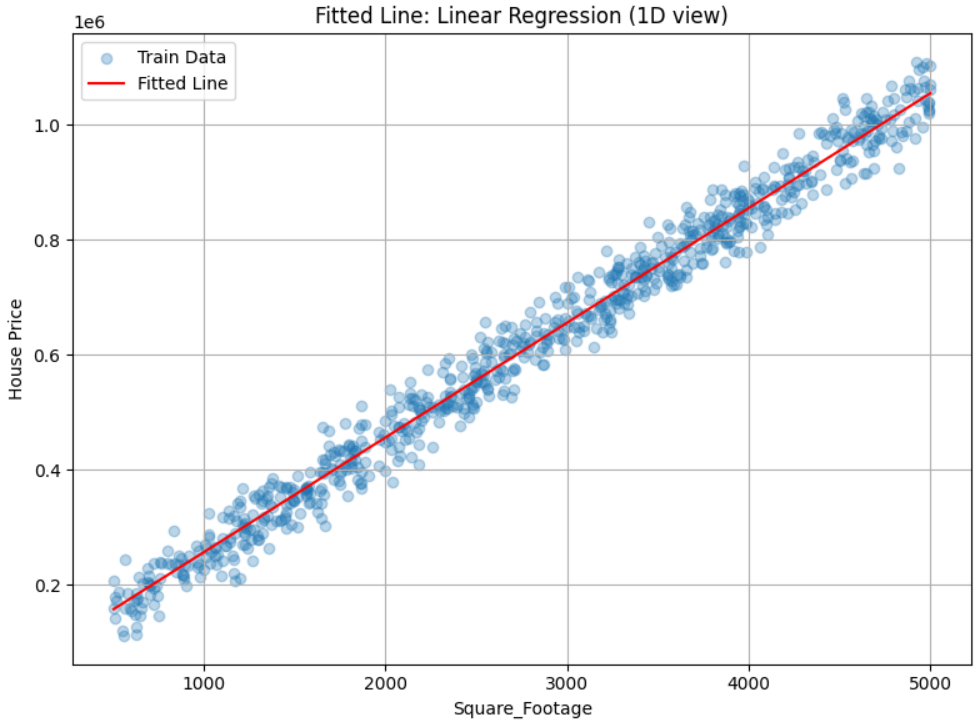

Figure 4: Fitted Line

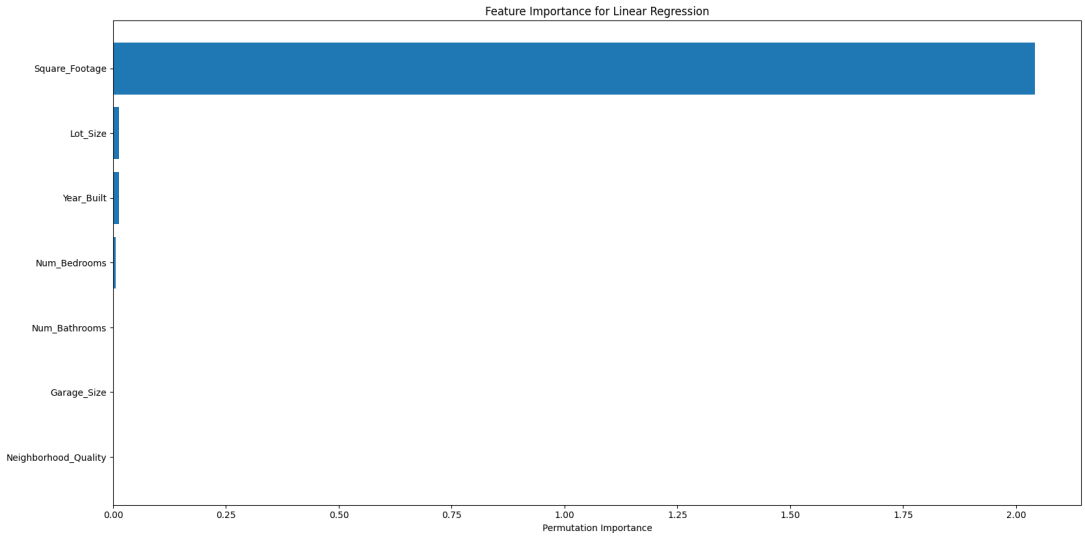

Figure 5: Features Importance